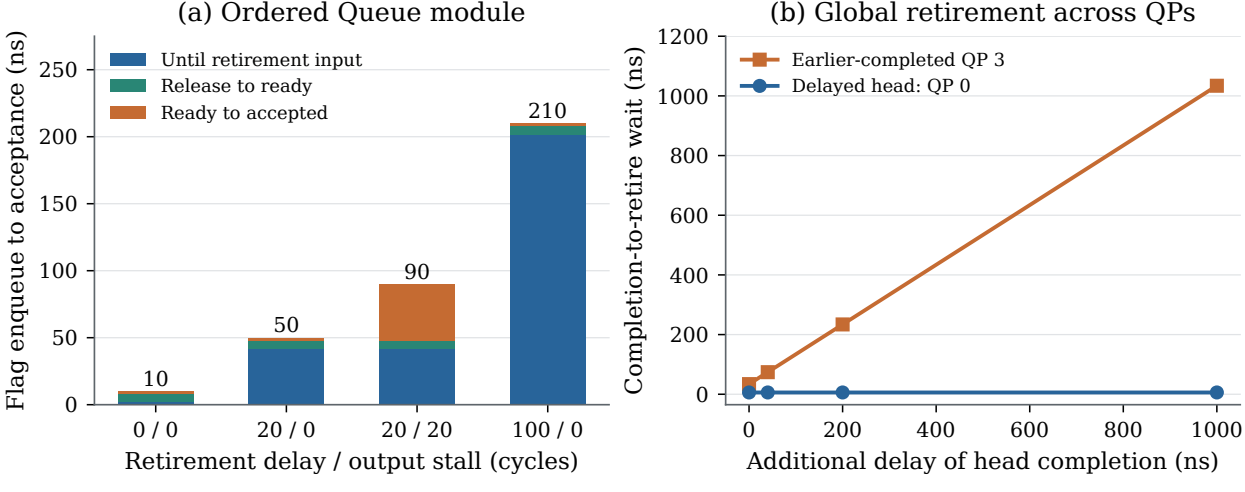
(a) Ordered Queue module
Until retirement input
Release to ready
Ready to accepted
Flag enqueue to acceptance (ns)
0
50
100
150
200
250
10
50
90
210
0 / 0
20 / 0
20 / 20
100 / 0
Retirement delay / output stall (cycles)
(b) Global retirement across QPs
Earlier-completed QP 3
Delayed head: QP 0
Completion-to-retire wait (ns)
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
Additional delay of head completion (ns)